

Alif Jawad

Tucson, Arizona, United States

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RESEARCH PROFILE

Ph.D. student in Mechanical Engineering with research experience in Flash Joule Heating synthesis of MAX phases, MXene-related materials, electrochemical materials, materials characterization, molecular dynamics simulations, and machine-learning-assisted materials modeling. Current research focuses on rapid synthesis pathways for low-entropy and high-entropy MAX phases, molten-salt-assisted processing, and structure-property relationships in advanced functional materials.

EDUCATION

University of Arizona

Ph.D. Student in Mechanical Engineering

2024 – Present

Tucson, Arizona, United States

- **Advisor:** Dr. Majid Beidaghi
- **Current GPA:** 3.75/4.00
- **Research Focus:** Flash Joule Heating synthesis of MAX phases, MXene-related materials, electrochemical materials, and materials characterization.

Bangladesh University of Engineering and Technology

B.Sc. in Mechanical Engineering

2018 – 2023

Dhaka, Bangladesh

- **GPA:** 3.53/4.00
- **Undergraduate Thesis:** Analysis of tensile strength of high-entropy alloy FeCrCuCoNi using molecular dynamics simulations and machine learning.

RESEARCH INTERESTS

- MXenes and MAX phases
- Flash Joule Heating synthesis
- High-entropy ceramics
- 2D materials
- Electrochemical energy storage
- Materials characterization
- Molecular dynamics simulations
- Machine learning for materials science

RESEARCH EXPERIENCE

Ph.D. Researcher, Beidaghi Electrochemistry Research Group

Supervisor: Dr. Majid Beidaghi

2024 – Present

University of Arizona

- Developing and optimizing Flash Joule Heating synthesis routes for Ti-, Nb-, and V-based MAX phases, with emphasis on precursor composition, reaction temperature profile, dwell condition, and phase purity.
- Investigating synthesis pathways from low-entropy to high-entropy MAX phases using rapid thermal processing and molten-salt-assisted reaction environments.
- Performing structural and chemical characterization of synthesized materials using X-ray diffraction, scanning electron microscopy, energy-dispersive spectroscopy, Raman spectroscopy, and related techniques.
- Analyzing phase evolution, impurity formation, and microstructural differences between Flash Joule Heating and conventional furnace synthesis routes.
- Collaborating on electrochemical evaluation of silicon–MXene ink electrodes for energy-storage applications.
- Collaborating on the preparation of delaminated MXene films with different intercalants for membrane and filtration-related applications.

- Conducted molecular dynamics simulations to calculate tensile strength of FeCrCuCoNi high-entropy alloys under varying compositional conditions.
- Generated low-fidelity single-crystal and high-fidelity polycrystalline simulation datasets for machine-learning-assisted mechanical property prediction.
- Trained a multi-fidelity physics-informed neural network model to predict tensile strength from composition-dependent molecular dynamics data.
- Analyzed simulation errors and fidelity relationships to improve predictive performance for high-entropy alloy strength estimation.

PUBLICATIONS

Journal Article

2025

N. E. Chowdhury, A. **Jawad**, A. Rahman, and M. J. A. Khan

- “Multi-fidelity neural network–based prediction of tensile strength of high-entropy alloy FeNiCoCrCu using molecular dynamics data.” *Journal of Molecular Modeling*, 31(8), 1–9.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Fundamentals of Materials for Engineers

Fall 2024 – Spring 2026

AME/MSE 331R, Course Instructor: Dr. Majid Beidaghi

University of Arizona

- Graded homework, course projects, and final assessments for multiple offerings of an undergraduate materials science course.
- Led problem-solving sessions and provided office-hour support for students on topics including crystal structures, phase diagrams, mechanical properties, and materials processing.
- Supported students in completing final course projects by providing technical guidance, feedback, and instructional resources.
- Prepared recorded instructional materials to assist students with project expectations and course problem-solving strategies.

TECHNICAL SKILLS

Materials Characterization: XRD, SEM, AFM, EDS, Raman Spectroscopy, UV-Vis Spectroscopy, TGA

Electrochemical Characterization: Cyclic Voltammetry, Galvanostatic Charge-Discharge, Electrochemical Impedance Spectroscopy

Synthesis and Processing: Flash Joule Heating, solid-state synthesis, molten salt processing, glovebox operation, fume hood operation, centrifugation, FlackTek SpeedMixer

Simulation and Modeling: LAMMPS, MATLAB, ANSYS Workbench, molecular dynamics simulations, machine learning, deep learning

Programming: Python, C, Arduino

Data Analysis and Visualization: Origin, OVITO, Tecplot 360, Excel, PowerPoint, LaTeX

CAD and Design: SolidWorks, AutoCAD

PROFESSIONAL EXPERIENCE

Industrial Attachment Trainee

2022

Unilever Bangladesh Limited

Chittagong, Bangladesh

- Calculated cooling load requirements for a production floor and incorporated measured dimensions and calculations into existing technical drafts.
- Tabulated process water-loss data and supported optimization efforts for water-use reduction in industrial operations.

LEADERSHIP AND SERVICE

Public Relations Officer
Bangladesh Students Association

2025 – 2026
University of Arizona

- Organized student association events and coordinated activities for the Bangladeshi student community at the University of Arizona.
- Managed outreach and communication activities to increase engagement and participation in association programs.

Team Captain
Hall Volleyball, Badminton, Basketball, and Table-Tennis Teams

2022
BUET

- Organized and managed multiple hall sports teams, leading several teams to semi-final placements in university-level tournaments.
- Coordinated team logistics and created hall funds to support player development and team activities.

Top 140 National Participant, Ennovators 6.0
Business Case Competition by Banglalink

2022
Bangladesh

- Selected among the top 140 national participants after several rounds of a competitive business case competition.

REFERENCES

References available upon request.